

PERSONAL DETAILS

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EDUCATION

Doctor of Philosophy (PhD), Computer Science

September 2018 -
January 2026

Northwestern University, Evanston, IL

GPA: 3.75/4.0

THILT lab

Research in Machine Learning

Bachelors of Science, Computer Science

June 2014 - May 2018

University of Maryland, Baltimore County, Baltimore, MD

GPA: 3.54/4.0

CMSC GPA: 3.8

Minor in Biology

EMPLOYMENT

Graduate Research Assistant

September 2018 - January 2026

Northwestern University - THILT Lab

PI: Dr. Marcelo Worsley

Designed and developed a Multimodal Learning Analytics tool, BLINC, that leverages machine learning to provide real time and post hoc feedback on college students' collaboration skills from audio and video data. Coordinated a regularly changing development team of 4-8 undergraduates and graduate students. Tested multiple hypotheses through user studies and A/B testing to refine ethical practices around user preferences and to test the best ways to present collaborative analytics to users to maximize collaborative skills and literacy improvement. Applied concepts of multimodal analytics to perform multimodal fusion and handle issues with temporal alignment between modalities. Integrated, refined, and combined common machine learning methods, Google Cloud Processing, HuggingFace models, Computer Vision, LLMs, and RAG to enable the quantification and development of student collaboration skills over short (<10 min) and long (>60 min) collaborative sessions. Orchestrated end-to-end ML AWS/NGINX infrastructure integrating a React frontend with specialized Python ML servers, supporting over 30 users and averaging less than 1s latency.

Data Science Research Intern

June 2018 - August 2018

Adobe

Supervisor: Dr. Tak Yeon Lee

Designed and presented a proof-of-concept demo of quickly (<10s) recommending products in Augmented Reality, using ARKit and ARCore, by engineering an object classification pipeline using Python OpenCV to extract key room features.

Undergraduate Research Assistant

July 2016 - May 2018

University of Maryland, Baltimore County

Supervisor: Dr. Marie desJardins

Developed multiagent models using POMDPs to improve the performance and feasibility of automated agent navigation across discrete and continuous domains.

STEM Instructor and Mobile App Expert

August 2016 - May 2018

Howard County Library System

Taught 6th - 12th grade students Computer Science, Math, Physics, and other STEM subjects. Also, aided in developing an app for the library system to help students study while out of the library and school.

Summer Undergraduate Research Fellow

June 2017 - August 2017

University of Arizona

Supervisor: Dr. Larry Head

Worked on simulating a Vehicular Ad-hoc Network(VANET) with different configurations of the physical layer to compare the performance of the currently licensed 5.9 GHz band to the unlicensed analog TV white space.

Mobile and Connected Devices Sales Consultant

October 2015 - August 2016

Best Buy

Assisted and sold connected devices such as tablets, security camera systems, routers, headphones, and speakers by learning extensive details about the products.

Summer Undergraduate Research Fellow

May 2015 - August 2015

National Institute of Standards and Technology

Supervisor: Dr. Martin Burns

Implemented the Green Button Standard on the NIST campus and wrote documentation for the implementation of the Green Button interface by creating a messaging pathway that allowed MySQL to communicate with Python and Python to communicate with Java while creating xml files that the Green Button server can process.

STEM Teaching Assistant

October 2013 - August 2015

Howard County Library System

Assist teachers at the library system with 6th - 12th grade Computer Science, Math, Physics, and other STEM subjects.

HONORS & AWARDS

- GEM Fellow

VOLUNTEER SERVICES

College Gardens of Arbutus - 2014

Help elementary students with their homework and other assignments and serve as a role model in a western Baltimore community

ACADEMIC SERVICES

Journal Reviewer

European Association for Research on Learning and Instruction

Learning and instruction

2025

Teaching Assistant

Northwestern University

Into to Artificial Intelligence - CS 348

2025

Teaching Assistant

Northwestern University

Inclusive Making - CS 311

2024

Teaching Assistant

Northwestern University

Human Computer Interaction - CS 330

2022

Teaching Assistant

University of Maryland, Baltimore County

Data Structures - CMSC 341

2017

Tutor

University of Maryland, Baltimore County

Introduction to Biology - BIOL 121 and 122

2015

PUBLICATIONS

Anderson, Khalil (2026). "Designing a Multimodal Collaboration Analytics Tool to Support Students' Collaboration Literacy Development". PhD thesis. Northwestern University.

Eze, Tochukwu, **Anderson, Khalil**, and Marcelo Worsley (2026). “Evaluating AI-based Visual Privacy through Cartoonification: Impact on Expressive Participation and Consent for Data Retention”. In: *Computers and Education Open*.

Eze, Tochukwu, Lucas Zurbuchen, **Anderson, Khalil**, and Marcelo Worsley (2026). “A collaboration literacy analytics framework (CLAF): Investigating students’ perspectives on collaboration quality indicators and feedback utilization”. In: *International Journal of Computer-Supported Collaborative Learning*.

Xie, Dawei, **Anderson, Khalil**, Tochukwu Eze, Chenghong Lin, Bookyung Shin, and Marcelo Worsley (2026). “CLARA: An AI-Augmented Analytics Dashboard for Collaboration Literacy”. In: *International Conference on AI in Education*.

Anderson, Khalil (2022). “Real-time Feedback for Developing Conversation Literacy”. In: *International Conference on Multimodal Interaction*.

Worsley, Marcelo, **Anderson, Khalil**, Natalie Melo, and Jang JooYoung (2021). “Designing Analytics for Collaboration Literacy and Student Empowerment”. In: *Journal of Learning Analytics*.

Anderson, Khalil, Theodore Dubiel, Kenji Tanaka, and Marcelo Worsley (2019). “Chemistry Pods: A Mutlimodal Real Time and Retrospective Tool for the Classroom”. In: *International Conference on Multimodal Interaction*.

Anderson, Khalil, Lauren Lusk, Marti Hands, and Garrett Vanhoy (2017). “Validation of a CRV Model Using TVWS Measurements”. In: *WinnComm*.

SKILLS

Python	Expert
Java	Expert
NLP	Expert
MySQL	Advanced
C++	Advanced
CSS	Advanced
HTML	Advanced
Android	Advanced
iOS	Advanced
ReactJS	Advanced
Angular	Advanced
AWS	Advanced
Google Cloud Services	Advanced
Prompt Engineering	Advanced
User and A/B Testing	Advanced
Computer Vision	Advanced
Retrieval Augmented Generation (RAG)	Advanced
LLMs (Gemini, Claude, and ChatGPT)	Advanced
C	Proficient
Assembly (x86, ARM)	Proficient
Scheme	Proficient
PHP	Proficient

REFERENCES

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